Formal-Informal Linkages in Indian Textiles & Apparel

Dissertation Progress Report — Stage 1 Analysis

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Dissertation Progress Presentation

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Outline

1. Research Questions & Theoretical Framework
2. Data & Methodology
3. Sample & Descriptive Statistics
4. Regression Results
5. Limitations & Next Steps
6. Summary

Research Questions

Core Hypothesis

Formal sector productivity **spills over** to the informal sector through two channels: informal wages and strategic outsourcing (job work).

RQ1: Wage Pass-Through

Does higher formal sector productivity raise informal sector wages?

$$\ln w_{si}^{inf} = \alpha + \beta \ln A_{si}^{f} + \varepsilon_{si}$$

RQ2: Strategic Outsourcing

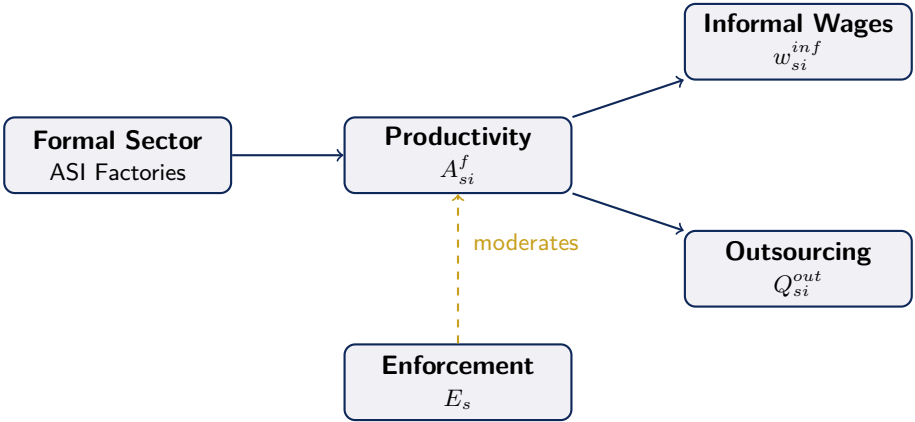
Does formal sector productivity drive outsourcing intensity to informal job-work firms?

$$\ln Q_{si}^{out} = \alpha + \gamma \ln A_{si}^{f} + \varepsilon_{si}$$

RQ3: Enforcement Moderation

Does labor law enforcement E_s moderate these linkages across states?

Theoretical Mechanism



Context: India's Textiles (NIC-13) & Apparel (NIC-14) sectors are deeply integrated — large formal factories outsource labour-intensive tasks to millions of small informal "job work" enterprises.

Data Sources

Formal Sector — ASI 2023–24

Annual Survey of Industries

- **68,641** total factories
- **8,137** textiles & apparel factories
- 33 states with usable data
- Key variables: output (Block J), employment (Block E), outsourcing cost (Block F)

Source: MoSPI / CSO

Informal Sector — ASUSE 2023–24

Annual Survey of Unincorporated Sector Enterprises

- **144,910** unique enterprises
- **7,519** job-work enterprises (textiles & apparel)
- Key filter: `contract_manuf_service=1`
- Key variables: hired workers (Item 511), wages (Item 559)

Source: MoSPI / NSS

Enforcement Data — E_s

Inspections-to-factories ratio per state. Source: Ministry of Labour & Employment Annual Report.

E_s = Total Inspections/Working Factories Range: $[0, 1]$, Mean = 0.22, SD = 0.196

Key Methodological Choices (and Why)

Analysis Level: State × Industry

- **Original plan:** District–Industry
- **Constraint:** Public ASI data suppresses all district/state geo-codes for confidentiality
- **Solution:** Use state codes from ASI field `a7`; ASUSE `district` field maps to states
- **Justification:** Enforcement is a state-level policy — state analysis is theoretically natural

Data Limitation (Documented)

- No district-level analysis possible from public data
- MOSPI restricted-access data would enable district IDs (3–6 month wait)
- This limitation narrows to N=47 obs

Job Work Definition

`contract_manuf_service = 1`

$\Rightarrow$ 7,519 enterprises
(vs. only 30 using `nature_of_operation=3`)

Variable Construction

Variable	Construction	Source Field
Formal Sector (ASI)		
A^f_{si} — Productivity	Output / Employment	J113 / (EI7/300)
Q^{out}_{si} — Outsourcing	Contract cost / Output	F7 / J113
L^f_{si} — Employment	$\sum(\text{mandays})/300$	EI7
Informal Sector (ASUSE)		
w^{inf}_{si} — Avg. Wage	Total wages / Hired workers	Item 559 / Item 511
Enforcement		
E_s — Enforcement	Inspections / Working factories	MoLE Annual Report
Aggregation		
State–Industry	Weighted mean (multiplier weights)	ASI: `mult`; ASUSE: `mlt`
Winsorization	1% – 99%	All log-transformed variables

Regression Models

Model 1 — Wage Pass-Through with Enforcement Interaction

$$\ln w_{si}^{inf} = \alpha + \beta_1 E_s + \beta_2 \ln A_{si}^f + \beta_3 \left(E_s \times \ln A_{si}^f \right) + \mu_i + \varepsilon_{si}$$

Model 2 — Strategic Outsourcing with Enforcement

$$\ln Q_{si}^{out} = \alpha + \delta_1 E_s + \delta_2 E_s^2 + \delta_3 \ln A_{si}^f + \mu_i + \varepsilon_{si}$$

- μ_i = 2-digit NIC fixed effects (absorbs industry-level differences)
- Standard errors clustered at state level
- E_s^2 term tests for non-linear (inverted-U) enforcement effect

- **Sample**: 47 state–industry cells, 26 states, 2 industries
- **Industries**: NIC-13 (Textiles), NIC-14 (Apparel)
- **Estimator**: OLS with high-dim fixed effects (`fixest` package)

Sample Construction Summary

ASI: All factories	**68,641**
ASI: Textiles & Apparel	**8,137**
ASI: State–Industry cells	**58 cells, 33 states**
ASUSE: Job-work enterprises	**7,519**
Merged: ASI $\cap$ ASUSE cells	**57 cells**
Regression sample (with E_s)	**47 cells, 26 states**

Descriptive Statistics

Variable	N	Mean	Median	Std. Dev.	Range
$\ln w_{si}^{inf}$ (Informal wage, log)	47	3.394	3.445	0.457	[2.33, 4.09]
$\ln A_{si}^{f}$ (Formal productivity, log)	47	13.85	14.00	0.895	[10.42, 15.44]
$\ln Q_{si}^{out}$ (Outsourcing share, log)	47	≈ 0	0	—	[0, 3.9×10^{-5}]
E_s (Enforcement ratio)	47	0.223	0.165	0.196	[0.00, 1.00]
N^{firms} (ASI factories per cell)	47	98.7	20	—	[1, 905]
N^{inf} (ASUSE enterprises per cell)	47	84.5	57	—	[1, 305]

Enforcement Variation

Low enforcement: Chandigarh (0), Uttarakhand (0.018), Bihar (0.014)
High enforcement: J&K (0.654), Karnataka (0.403), Gujarat (0.477)
Outlier: Manipur ($E_s = 1.0$)

Key Correlations

	$\ln A^f$	E_s
$\ln w^{inf}$	-0.032	-0.281
$\ln Q^{out}$	$\mathbf{+0.285}$	-0.011

Main Regression Results

	Model 1 $\ln w_{si}^{inf}$	**Model 2** $\ln Q_{si}^{out}$
E_s	0.114	6.75×10^{-6}
	(5.087)	(1.39×10^{-5})
$\ln A_{si}^{f}$	-0.029	3.72×10^{-6} *
	(0.072)	**(1.74×10^{-6})**
$E_s \times \ln A_{si}^{f}$	-0.059	—
	(0.378)	
E_s^2	—	-4.26×10^{-6}
		(1.25×10^{-5})
NIC Fixed Effects	**Yes**	**Yes**
S.E. Clustered (State)	**Yes**	**Yes**
Observations	47	47
Within R^2	0.081	**0.102**

* $p < 0.05$; Standard errors in parentheses. Grey = not significant.

Interpreting the Results

Model 1 — Wage Pass-Through *(Not Significant)*

- No significant positive pass-through of formal productivity to informal wages
- Enforcement does not significantly moderate the wage channel ($p > 0.87$)
- **Why?** Small N (47 obs), cross-sectional data lack time variation
- **Direction**: Negative $\beta_2 = -0.029$ may reflect labour market segmentation

Model 2 — Outsourcing *(Significant!)* $p = 0.043$

- **Formal productivity significantly increases outsourcing intensity**
- $\hat{\delta}_3 = 3.72 \times 10^{-6}$, $t = 2.14$, $p = 0.043$
- Higher-productivity formal firms outsource *more* to informal job-work enterprises
- **Consistent with** value-chain specialisation hypothesis
- E_s terms not significant — enforcement doesn't deter outsourcing on its own

Enforcement & Informal Wages: Correlation Evidence

Raw Correlation: E_s & $\ln w^{inf}$

$$\text{corr}(E_s, \ln w^{inf}) = -0.281$$

- States with **stricter enforcement** tend to have **lower informal wages**
- Consistent with enforcement raising barriers for informal firms
- Informal firms pay less when they cannot absorb compliance costs
- Not causal — omitted state characteristics may explain this

State	E_s	$\ln w^{inf}$
Gujarat	0.477	∼3.1
Karnataka	0.403	∼3.3
Tamil Nadu	0.313	∼3.5
Haryana	0.103	∼3.7
Uttarakhand	0.018	∼4.1

Illustrative values from regression sample

Higher $E_s \downarrow$ waged

Limitations

Data & Geographic Constraints

1. **Small N**: Only 47 state–industry observations reduces statistical power
2. **Cross-sectional**: Single year (2023–24) — no panel variation
3. **ASI geography suppressed**: District codes unavailable in public unit-level data
4. **6 states missing** enforcement records (HP, UP, WB, Arunachal, Daman & Diu)

Methodological Notes

1. **Outsourcing proxy**: ASI `F7` captures contract costs but may miss informal subcontracting recorded off-books
2. **Job-work definition**: `contract_manuf_service=1` is broader than strict job work — all contract manufacturers included
3. **Weighted aggregation**: Multiplier-weighted means correct for sample design, but state cells vary in factory count (1 to 905)
4. **Manipur outlier**: $E_s = 1.0$ is an extreme case needing robustness check

Next Steps

Immediate (Before Final Submission)

1. **Robustness checks**
   - Winsorise E_s or use quartile dummies
   - Exclude Manipur ($E_s = 1.0$) outlier
   - Unweighted regression
2. **Multiple enforcement measures**
   - Inspectors per factory (not just inspections)
   - Besley–Burgess (2004) index as alternative
3. **Heterogeneity analysis**
   - Separate regressions for NIC-13 vs NIC-14
   - High- vs low-enforcement state splits

Longer Horizon

1. **Panel dimension**
   - Pool ASI/ASUSE 2022–23 and 2023–24 for two-period panel
   - State–industry FE would absorb more confounders
2. **Restricted-access data**
   - Apply to MOSPI for district-level ASI unit IDs
   - Would enable original district–industry design (N≈200)
3. **Stage 2 structural model**
   - Formalise the two-sector model
   - IV strategy for A^f endogeneity

Summary of Progress

What Has Been Done

- ✓ Processed ASI 2023–24 microdata (68k factories)
- ✓ Processed ASUSE 2023–24 microdata (145k enterprises)
- ✓ Identified job-work enterprises ($n = 7{,}519$)
- ✓ Constructed state–industry panel (47 obs)
- ✓ Collected & integrated state enforcement data (28 states)
- ✓ Estimated baseline regression models
- ✓ Significant result: productivity $\rightarrow$ outsourcing

Key Finding

Formal productivity

$\downarrow$

Significantly increases outsourcing to informal sector

$p = 0.043$

Consistent with value-chain specialisation theory

Pending

Wage pass-through channel not yet confirmed (insufficient N at state level)

Thank You

Dissertation Progress Report

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Data: ASI 2023–24, ASUSE 2023–24, MoLE Annual Report

Tools: R (`fixest`, `tidyverse`), LaTeX

Analysis level: State $\times$ 2-digit NIC Industry, $N = 47$

Questions & Feedback Welcome

Appendix: Variable Mapping (Verified)

Dataset	Block/Level	Field	Variable
ASI	Block A (ID)	`a7`	State code
ASI	Block A	`a5`	NIC-5 code
ASI	Block A	`mult`	Expansion multiplier
ASI	Block E (Emp.)	`EI7`	Total mandays worked
ASI	Block F (Expenses)	`F7`	Contract/outsourcing cost
ASI	Block J (Output)	`J113`	Total output value
ASUSE	Block 2 (ID)	`district`	State identifier
ASUSE	Block 2	`major_nic_5dig`	NIC-5 code
ASUSE	Block 2	`contract_manuf_service`	Job-work flag
ASUSE	Block 2	`mlt`	Expansion multiplier
ASUSE	Block 4 (Workers)	Item 511	Number of hired workers
ASUSE	Block 5 (Wages)	Item 559	Total wages paid (Rs)

Appendix: Enforcement Data by State

State	E_s	Inspections
Manipur	1.000	53
J&K	0.654	802
Tripura	0.589	389
Mizoram	0.375	3
Meghalaya	0.282	90
Karnataka	0.403	7,065
Gujarat	0.477	19,319
Odisha	0.349	1,023
Tamil Nadu	0.313	9,784

State	E_s	Inspections
Kerala	0.244	5,346
Telangana	0.260	4,623
Goa	0.197	147
Andhra Pradesh	0.156	2,982
Rajasthan	0.166	1,829
Maharashtra	0.123	4,745
Haryana	0.103	2,044
Punjab	0.095	1,998
Delhi	0.080	699
Chandigarh	0.000	0

Appendix: Detailed Regression Output (Model 1)

Model 1: $\ln w_{si}^{inf} \sim E_s \times \ln A_{si}^f$ | `NIC_2digit`, clustered SE

Variable	Estimate	Std. Error	t	p-value
E_s	0.11398	5.08678	0.022	0.982
$\ln A_{si}^f$	-0.02858	0.07154	-0.400	0.693
$E_s \times \ln A_{si}^f$	-0.05921	0.37841	-0.156	0.877
NIC FE	Yes (2 levels: NIC-13, NIC-14)			
Observations	47			
RMSE	0.4303			
Adj. R^2	0.010			
Within R^2	0.081			

Appendix: Detailed Regression Output (Model 2)

Model 2: $\ln Q_{si}^{out} \sim E_s + E_s^2 + \ln A_{si}^f$ | NIC_2digit, clustered SE

Variable	Estimate	Std. Error	t	p-value
E_s	6.75×10^{-6}	1.39×10^{-5}	0.487	0.631
E_s^2	-4.26×10^{-6}	1.25×10^{-5}	-0.340	0.737
$\ln A_{si}^f$	3.72×10^{-6}	1.74×10^{-6}	2.137	**0.043***
NIC FE	Yes (2 levels: NIC-13, NIC-14)			
Observations	47			
RMSE	9.649×10^{-6}			
Adj. R^2	0.038			
Within R^2	0.102			

* $p < 0.05$. S.E. clustered by state.